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GIZMo-Kria

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U.S. DEPARTMENT of ENERGY
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Agenda

- ✓ Hardware
- ✓ Firmware
- ✓ Software
- ✓ Environment Variables
- ✓ Hardware Setup
- ✓ System Startup

01

Hardware

Hardware

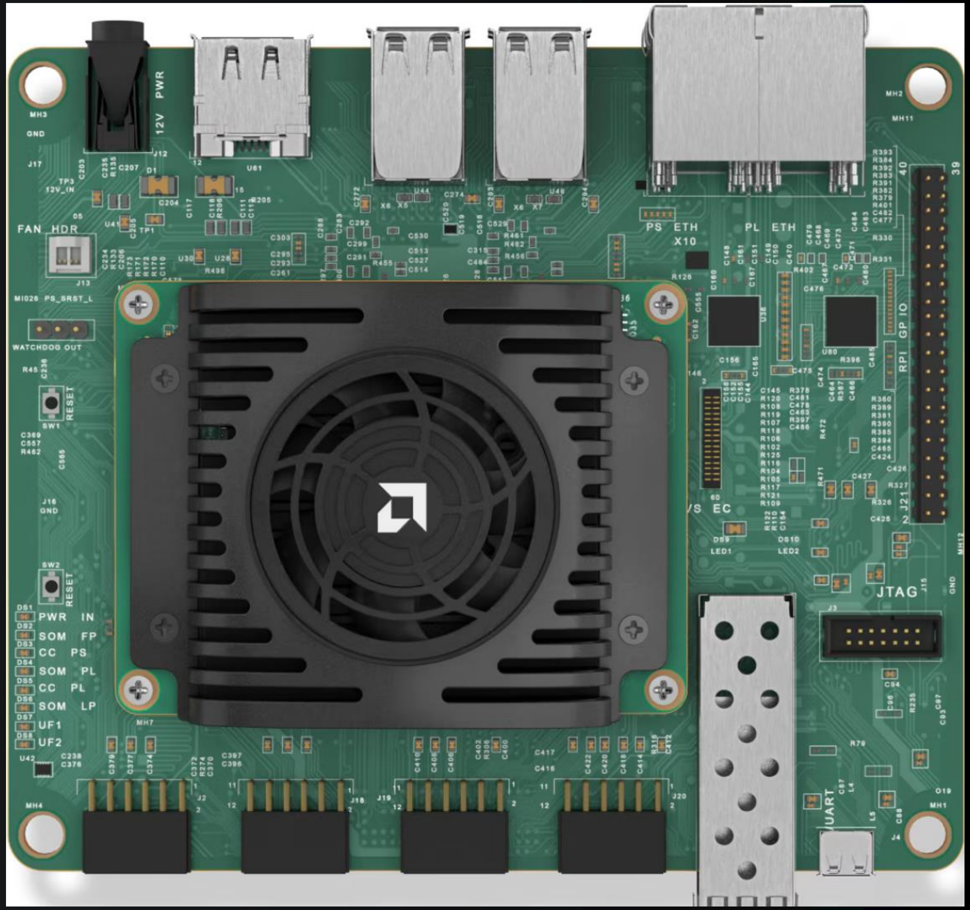
Overview

- Kria Robotics Kit
  - Custom FPGA Firmware Overlay
  - OS stored on SD Card
- Control Board
  - Interfaces Analog and Digital Hardware with I/O
- Audio Amplifier Board
  - Plays a track from an SD card when the system is triggered
- 5” Capacitive Touch Panel

Hardware

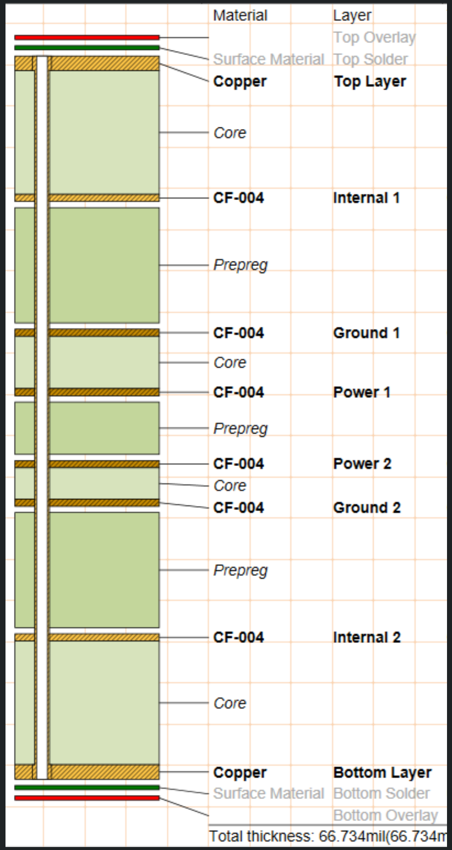
Kria Robotics Kit

- Rear I/O
  - Display Port
  - 4 x USB
  - 4 x GbE
    - 2 x PS
    - 2 x PL
- Internal I/O
  - 40-pin Raspberry-Pi Connector
  - JTAG
  - PL SFP
  - 4 x PMOD
  - UART/JTAG USB
- Inline 3A Fuse on the Power Input
- SD Card or Network Bootable

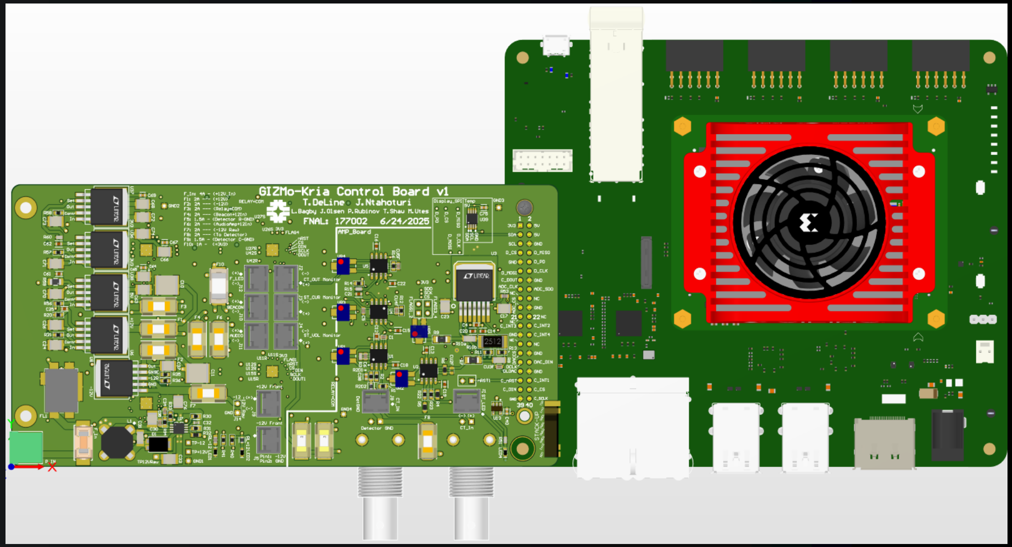


Hardware

Control Board



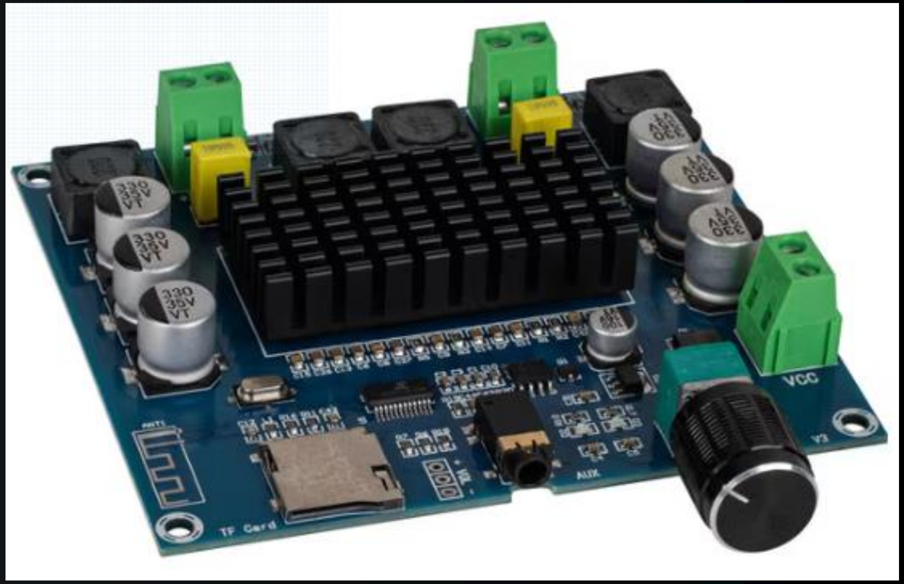
- 8 Layer PCB
  - 2 Ground Plane Layers
  - 2 Power Plane Layers
  - 4 Signal Layers
- I/O
  - 40-Pin Raspberry-Pi Connector
    - ADC (Connected to a Current Transformer)
    - DAC (Sent through the Detector GND to Building GND
    - SPI Display
    - Relay Drivers
    - Interrupts
  - BNC
    - Stimulus Output, CT Input
    - 2 Monitor BNCs for connecting an Oscilloscope
  - Latching Pin Header Connectors
    - LEDs / Lights
    - Audio Amplifier
- Fuses
  - All outputs are fused with replaceable fuses
  - Power inputs fused
    - 4A Control Board



Hardware

Audio Amplifier

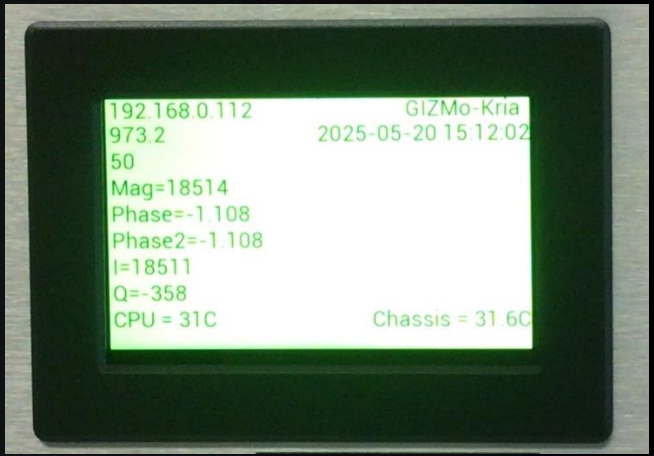
- Speakers
  - Stereo Output
  - 100W
- Power
  - 12V from control board relay
- Volume
  - Potentiometer adjustment
  - Volume Rocker Switch
- SD Card
  - Track plays on repeat when system is triggered
- 3.5mm Aux input
- Bluetooth

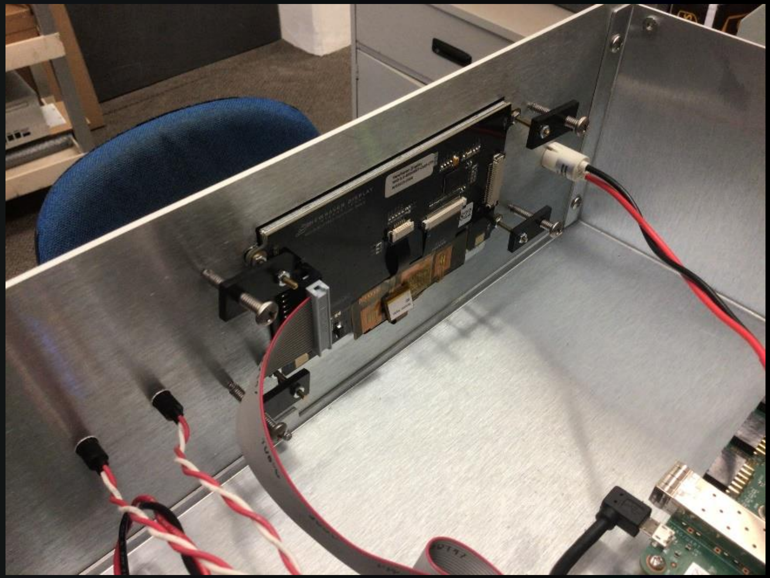


Hardware

Display

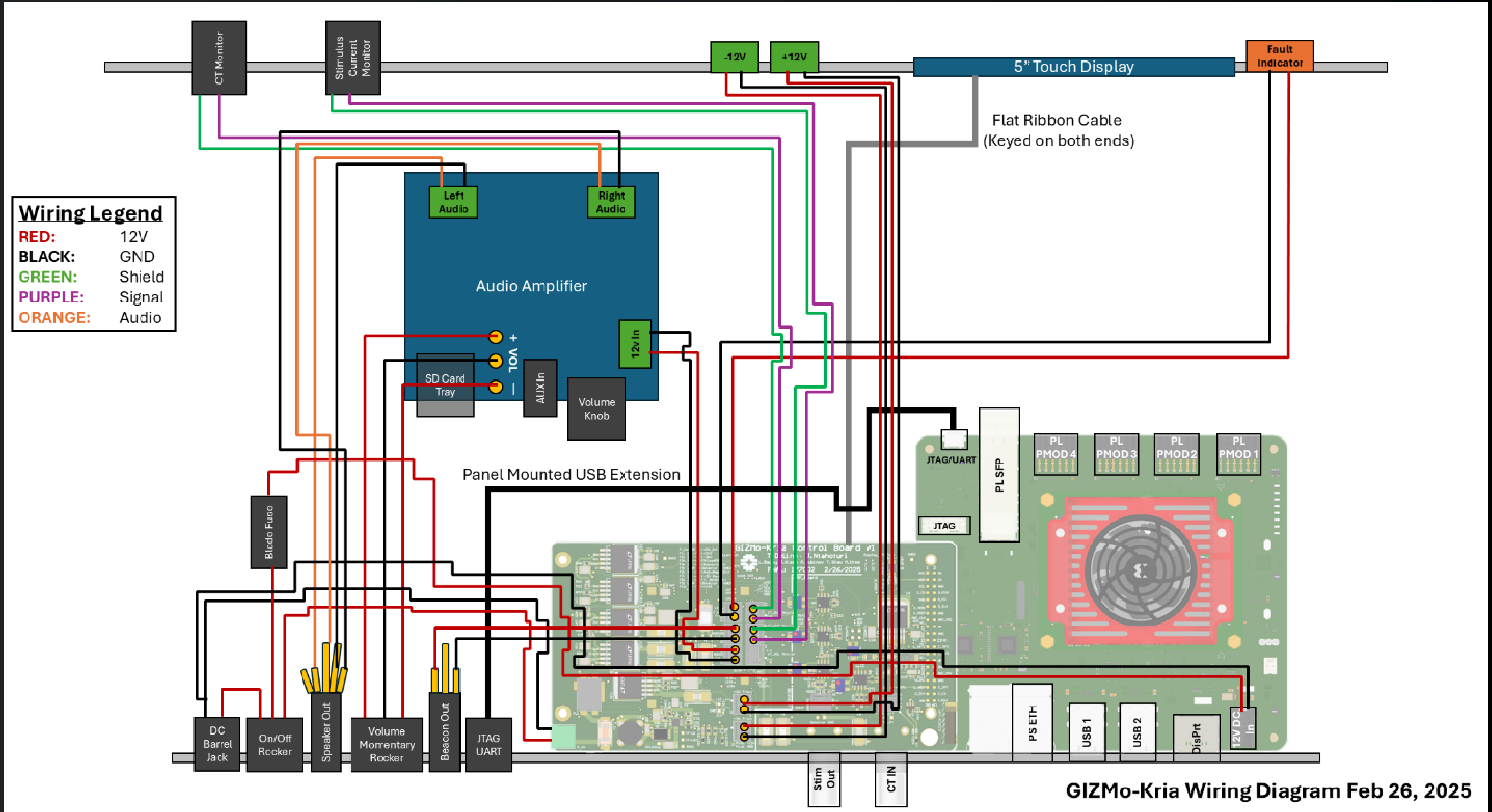
- New Haven Display
  - 5”
  - Capacitive Touch Panel
  - EVE Command Architecture
  - Power
    - 5V and 3v3 from Robotics Kit through 40-Pin header
  - EVE and ZMon binaries communicate using IPC with a socket setup at port 5055 to get data





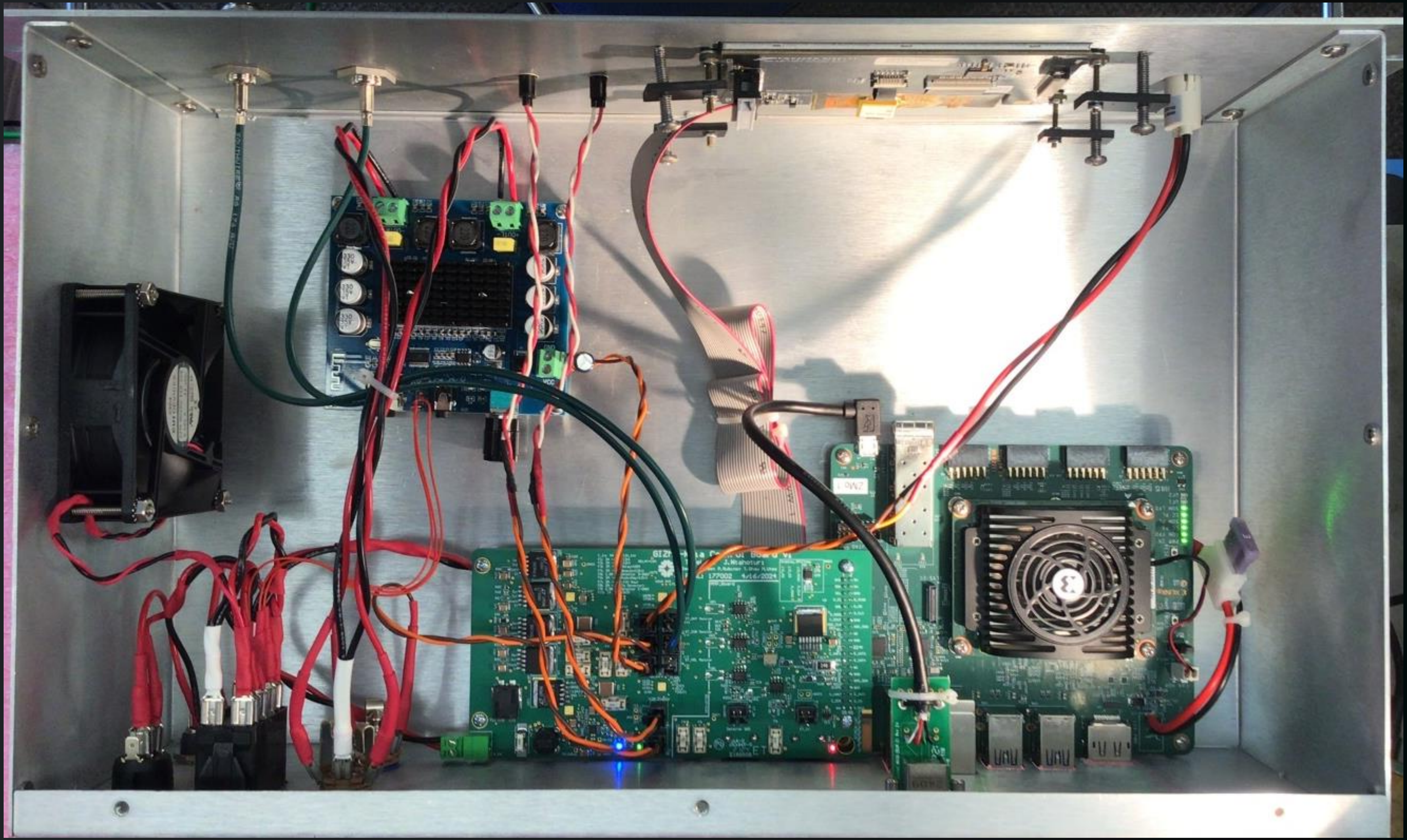
Hardware

Wiring Diagram



Hardware

Chassis Internal

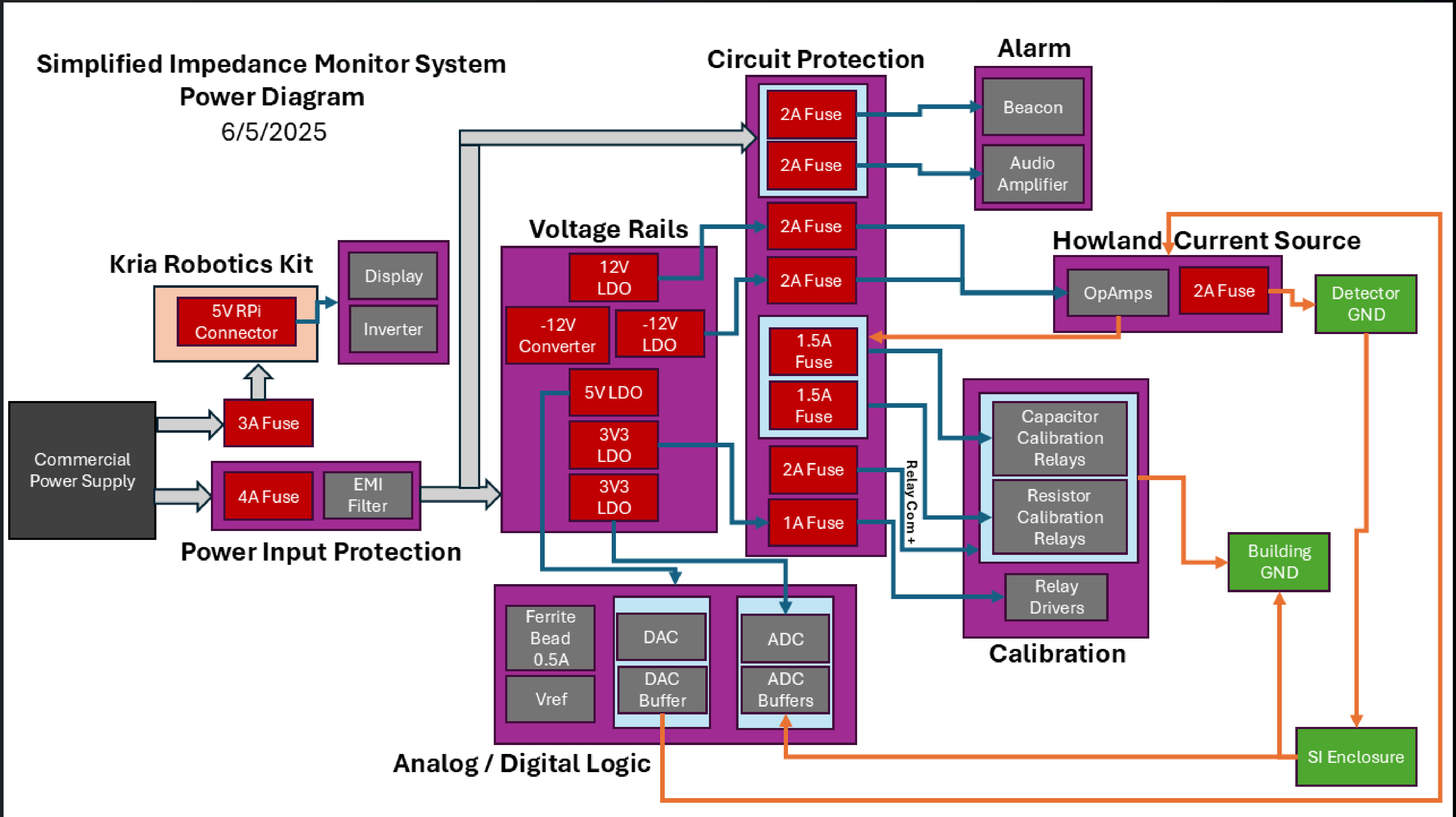


Hardware

Chassis Front and Rear Panels

Hardware

Power



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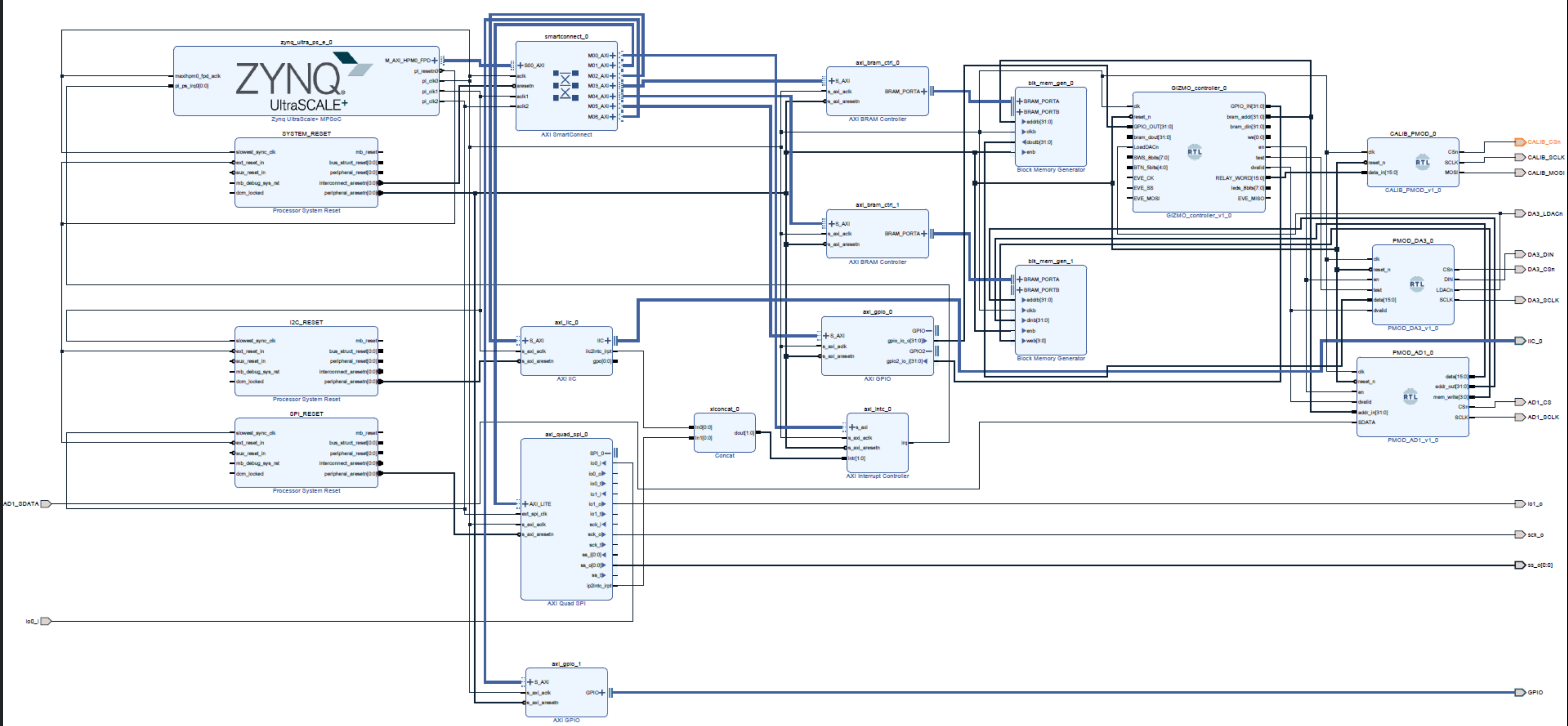
Firmware

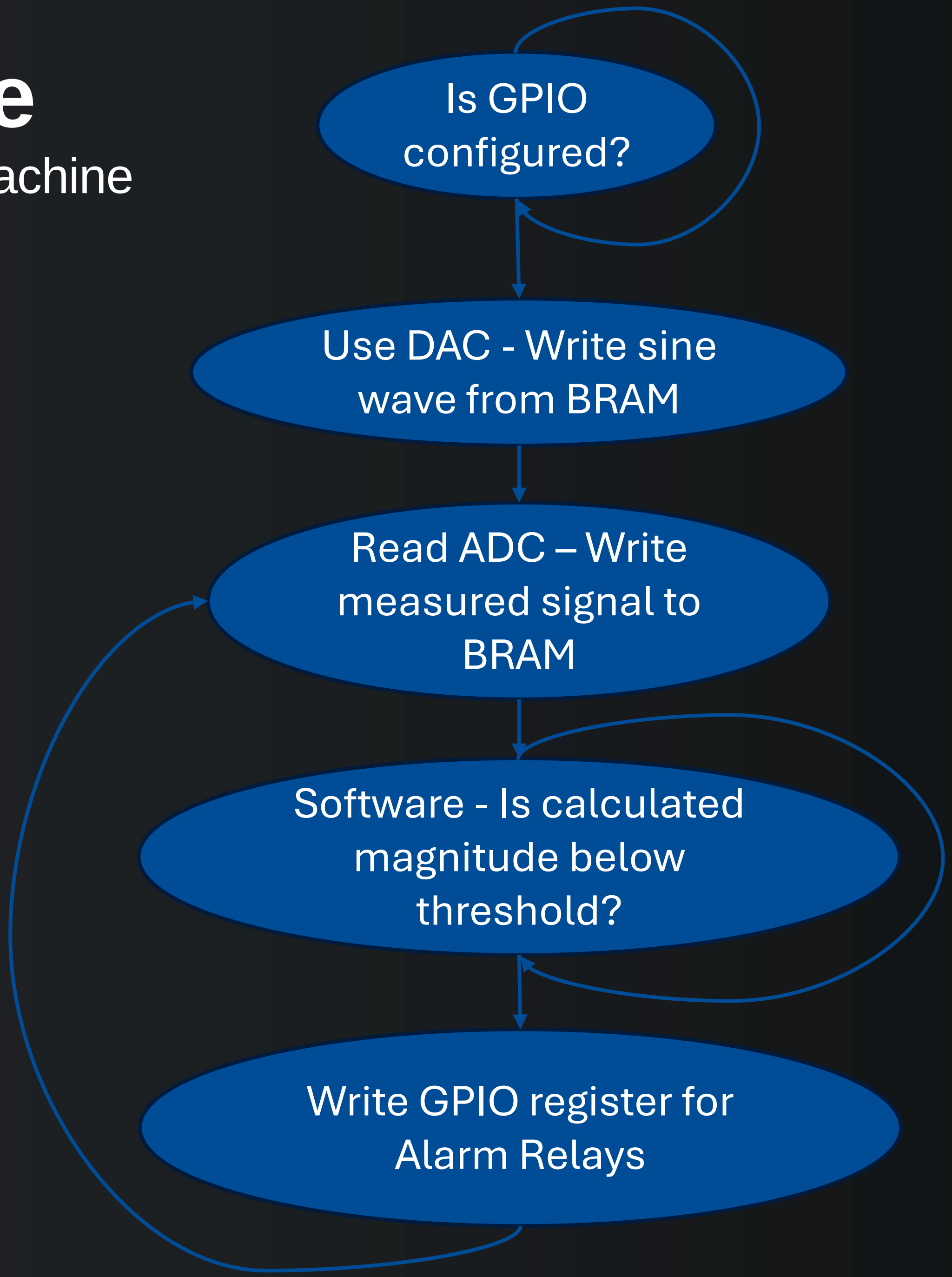
Firmware

System Block Diagram



Firmware

System State Machine



5” Display Panel functions independently of firmware state machine

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Software

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Software

Overview

- ZMon Binary
  - Commands: run, CAL, read_adc, set_rly
  - Server available at port 5055
    - Sends out measurement details to anyone that connects
- EVE SPI Display Binary
  - Writes to the SPI display using EVE architecture
  - Connects to port 5055 to get data from ZMon script and writes data to display
  - Reads i2c temperature sensor on the control board
- ZMQ Python Server
  - Available on port 5555
  - Commands: set_time (M or L), clear latch state, get data (Can be continuous), run, CAL, get_cal, set threshold, read_adc, get_adc, testHello.py
- OPC-UA Python Server Bridge
  - Available on port 4840
  - Receives commands from client, forwards the command to port 5555 for ZMQ Server to process.

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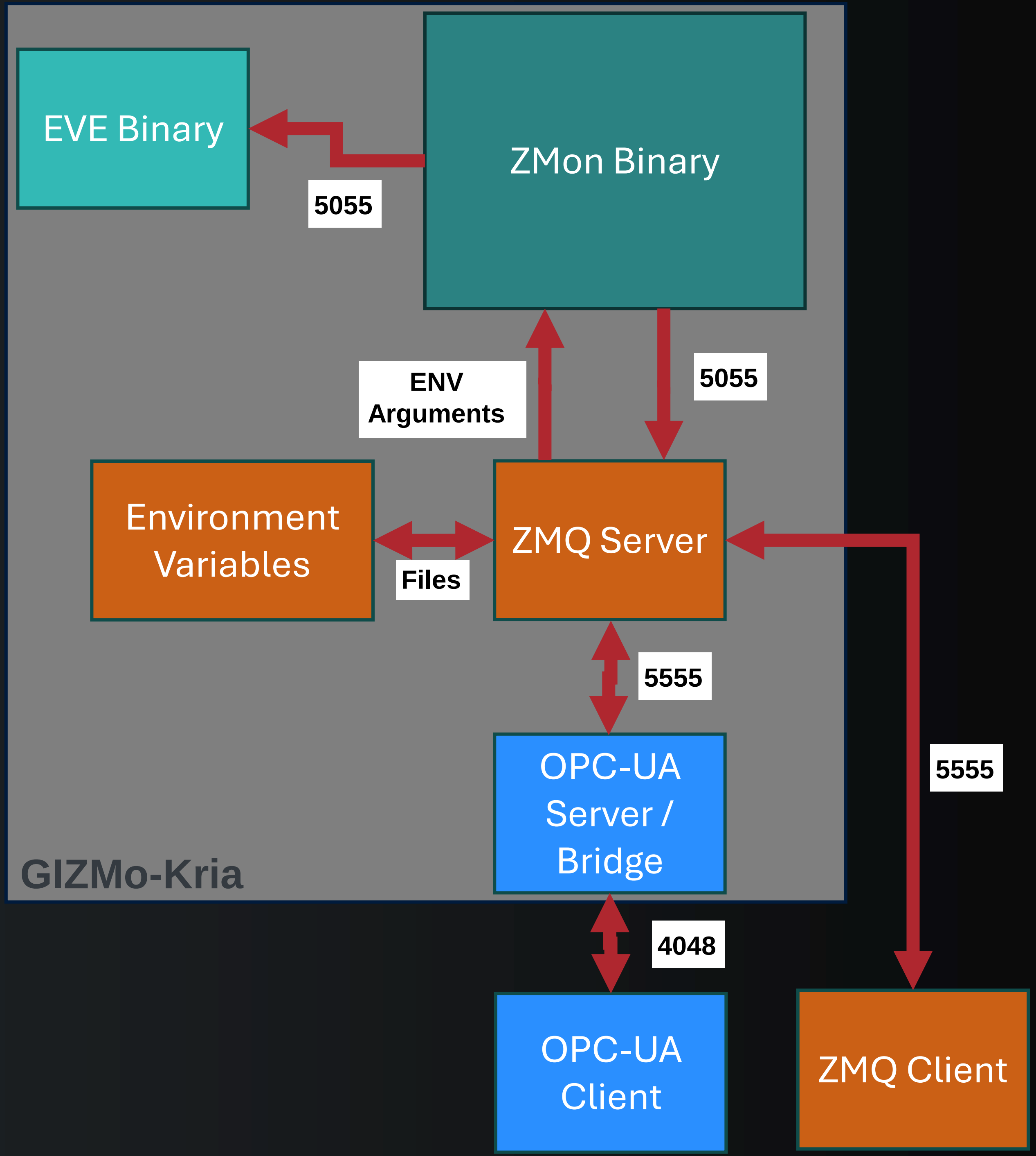
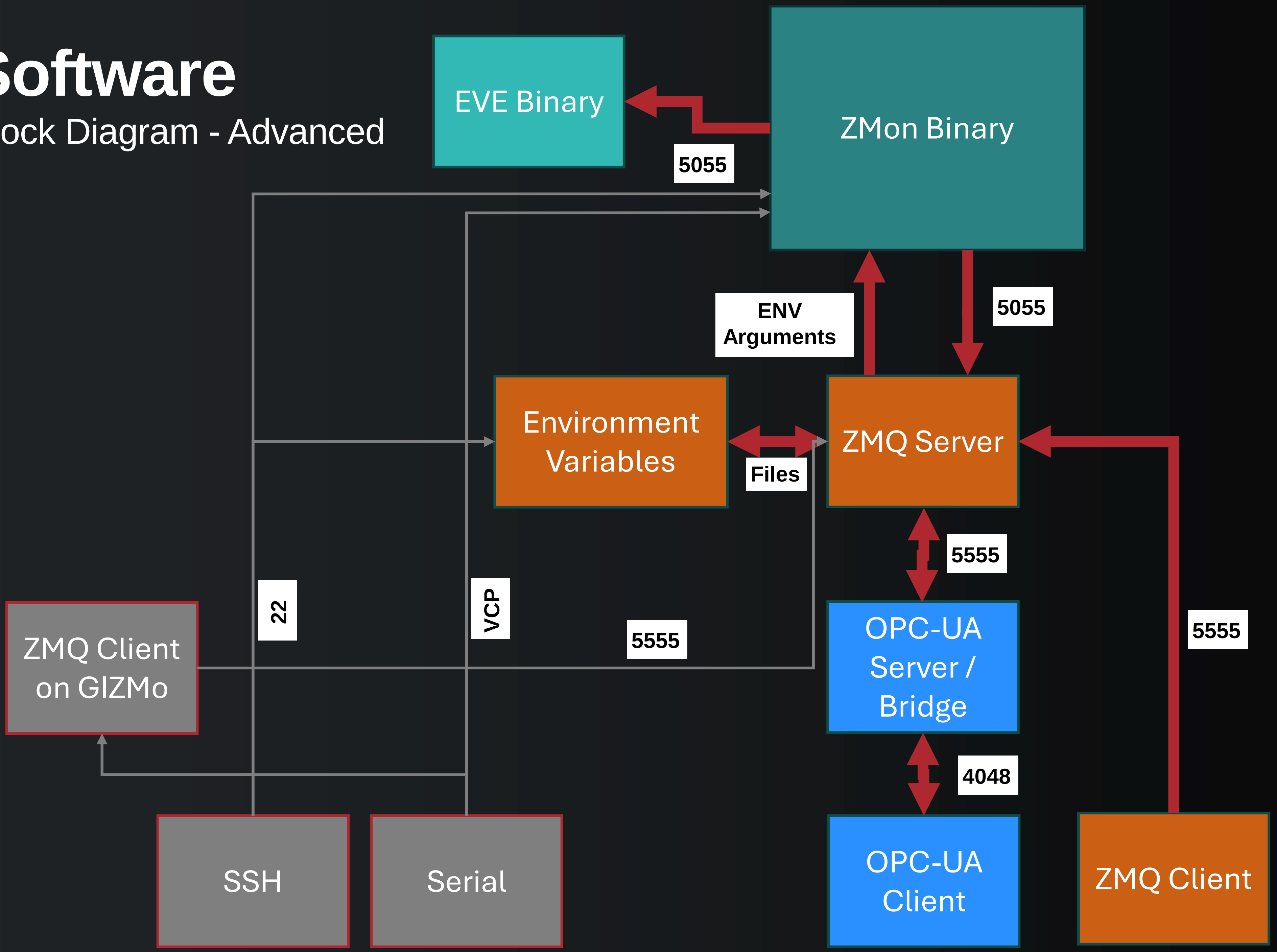
Software

Block Diagram



Software

Block Diagram - Advanced



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Environment Variables

Environment Variables

- ZMonArg1.env
  - ZMonArg1="set_th 10"
- ZMonArg2.env
  - ZMonArg2="run 1"
- ZMonArg3.env
  - ZMonArg3=""
- setRunInterval.env
  - export runInterval=1
- setThreshold.env
  - export threshold=10

05

Hardware Setup

Hardware Setup

Main Components

- Impedance Monitor
  - GIZMo-Kria
- Saturable Inductor Enclosure
  - Connects Detector Ground to Building Ground
- Speakers
  - 8-Ohm
  - 1/4“ Jack (Guitar/Audio Plug)
- Beacon
  - Powered from the Control Board inside of the Impedance Monitor Chassis
- BNC Cables

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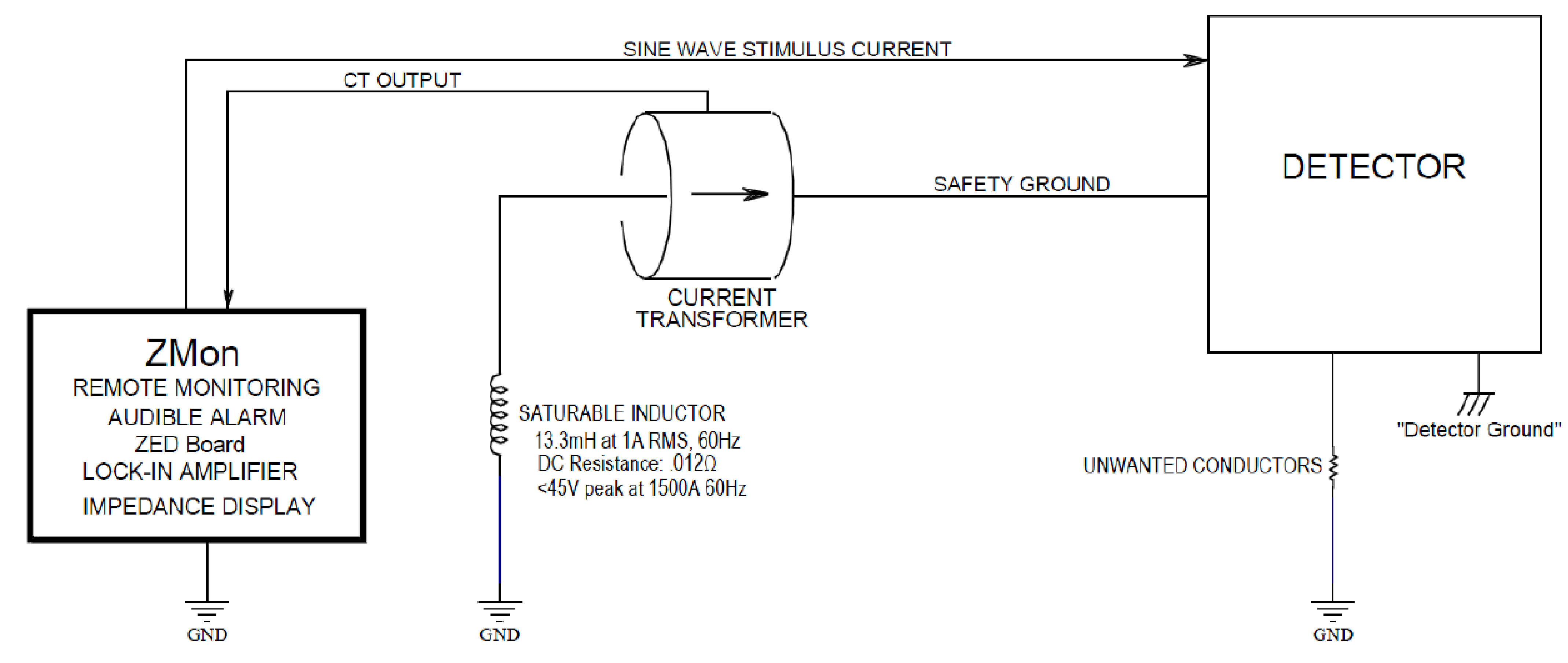
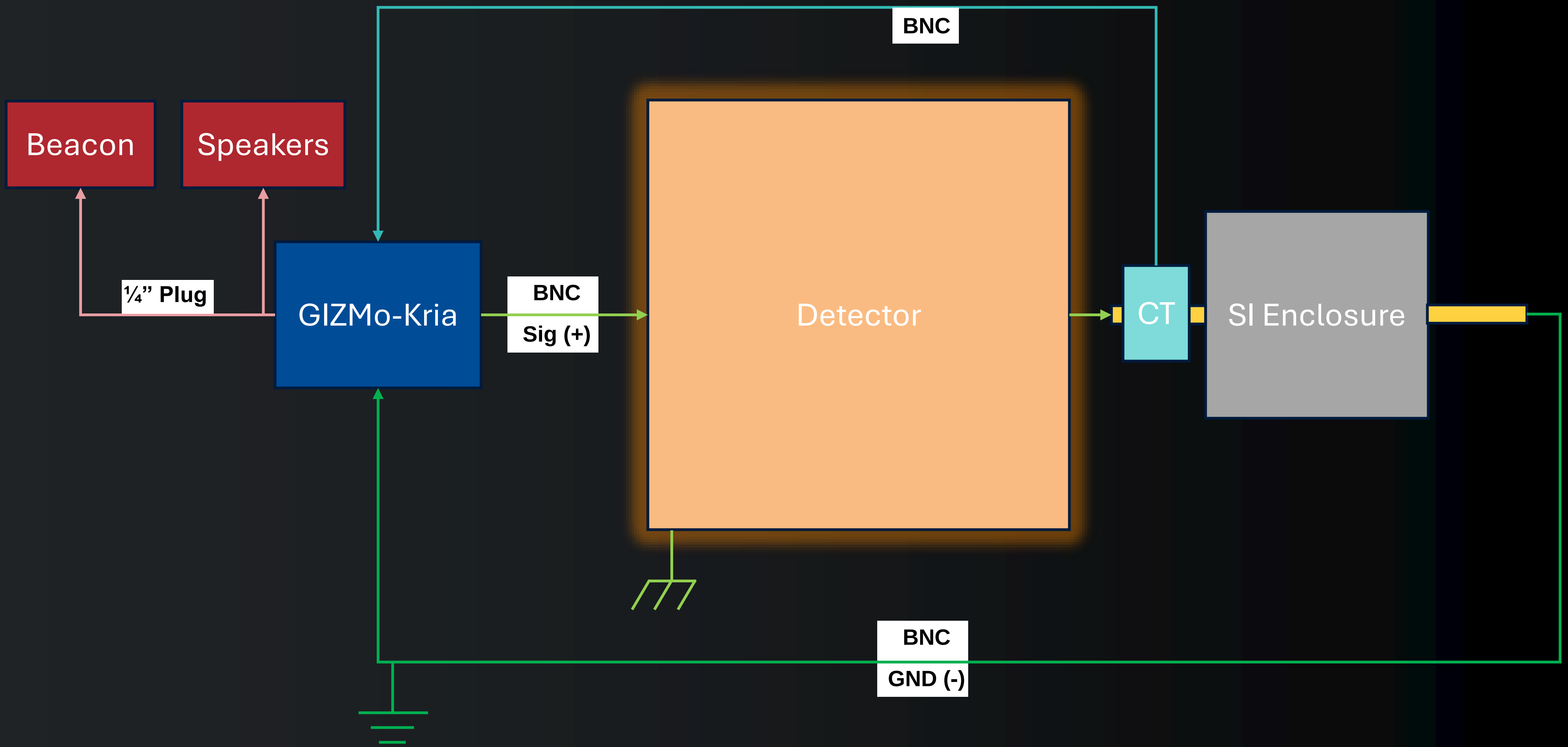
Hardware Setup

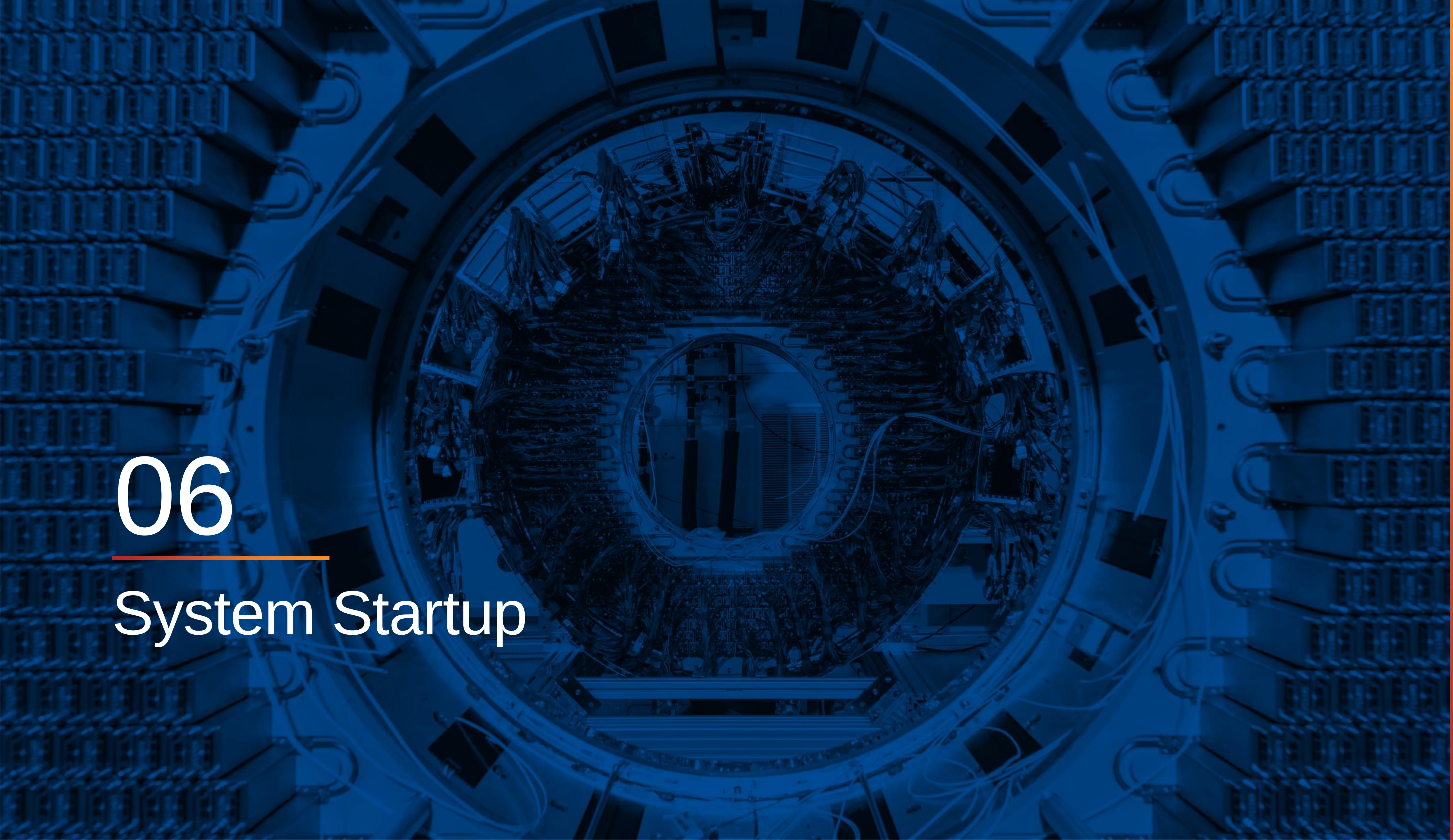
System Model



Hardware Setup

Block Diagram





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System Startup

System Startup

1. Connect Stimulus and CT cables to the detector, impedance monitor, and SI Enclosure with BNC cabling
2. Connect Beacon and Speaker output with ¼” Mono and Stereo Plugs
3. Connect Networking or Serial Cables
4. Connect Power and flip power switch
5. System auto loads:
    1. Firmware
    2. ZMon Binary
    3. Display Binary
    4. ZMQ Server
    5. OPC-UA Server
6. Run Calibration command from ZMQ or OPC-UA client
7. View results from client
8. Archive initial calibration run as detector baseline

07

G271 Bench Testing

G271 Bench Testing

1. FESHM9111 – GIZMo Team
2. ORC – Linda
3. Install Hardware
4. Add simulated short (Resistor and Capacitor Decade Boxes in Parallel with system)
5. Install library dependencies on Linux and Windows machine in G271
6. If no controls monitoring system in place, write a ZMQ or OPC-UA script to log data snapshots to Linux/Windows machine
7. Inspect system at regular intervals
8. Modify / Upgrade control software as necessary

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